

HumanMaximizer AI Lead Generation

Architecture & Design Document

Razor Infotech Pvt Ltd — AI Architect / GenAI Engineer Assignment

HumanMaximizer Lead Gen — Architecture

AI Lead Generation & Research Assistant · Razor Infotech Take-Home Assignment

System Architecture Diagram

```
flowchart TB
    subgraph CLIENT["CLIENT"]
        UI["REST Client · Swagger UI"]
    end

    subgraph API["API GATEWAY — FastAPI + Uvicorn"]
        direction LR
        EP1["POST /api/v1/search"]
        EP2["GET /api/v1/leads"]
        EP3["POST /api/v1/outreach/generate"]
        EP4["GET /api/v1/rag/query"]
        EP5["GET /health GET /metrics"]
    end

    subgraph PIPELINE["AGENT PIPELINE — LangGraph StateGraph · LeadState"]
        direction LR
        RA["ResearchAgent\n██████████\nSerpAPI search\nWeb scrape\nNDM extraction\nLLM summary"]
        QA["QualificationAgent\n██████████\n5-dim scoring 0-100\nCompleteness gate\nConditional edge"]
        SA["SalesAgent\n██████████\nRAG retrieval\nCold email gen\nLinkedIn gen\nSelf-critique"]

        RA -->|"enriched LeadState"| QA
        QA -- "score ≥ 35" --> SA
        QA -- "score < 35" --> DQ(["disqualified"])
        SA --> Q(["qualified + outreach"])
    end

    subgraph RAG["RAG PIPELINE — humanmaximizer.com knowledge base"]
        direction LR
        INGEST["WebIngestor\nAsync crawler · depth=3\n512-char chunks · overlap=64"]
        EMBED["nomic-embed-text\n768-dim · Ollama"]
        VDB["ChromaDB\nPersistentClient\nhnsw:cosine · 27 chunks"]
        INGEST --> EMBED --> VDB
    end

    subgraph LLM["LLM INFERENCE — Ollama"]
        MISTRAL["Mistral-7B-Instruct-v0.3 · Q4_K_M\n4-bit GGUF · ~4.4 GB VRAM · CPU/GPU"]
    end

    subgraph PERSIST["PERSISTENCE"]
```

```
DB[("SQLite → PostgreSQL\nSQLAlchemy async · Alembic")]\nend\n\nsubgraph OBS["OBSERVABILITY"]\n  direction LR\n  PROM["Prometheus /metrics\nstage_latency · lead_score\nrag_chunks · hallucination_checks"]\n  LS["LangSmith\nper-request agent trace"]\nend\n\nUI --> API\nEP1 --> PIPELINE\nEP3 --> SA\nEP4 --> VDB\n\nRA --> LLM\nQA --> LLM\nSA --> VDB\nSA --> LLM\n\nPIPELINE --> PERSIST\nPIPELINE --> OBS
```

Component Responsibilities

Component	Single Responsibility
FastAPI Gateway	Route validation, OpenAPI docs, middleware (metrics, CORS), DB session injection
LangGraph StateGraph	Immutable LeadState passing through nodes; conditional edge enforces qualification threshold
ResearchAgent	SerpAPI → top result → web scrape → DM regex extraction → Mistral-7B structured summary
QualificationAgent	5 deterministic scoring functions (0–20 each); completeness gate; LLM 2-sentence reasoning
SalesAgent	RAG query → cosine retrieval → Mistral-7B cold email + LinkedIn; self-critique post-pass
WebIngestor	One-time crawl of humanmaximizer.com; chunk → embed → upsert into ChromaDB
ChromaDB	Persistent cosine-similarity vector store; zero network dependency (PersistentClient)
Ollama / Mistral-7B	Single LLM endpoint shared across all 3 agents; Q4_K_M for 8 GB GPU / CPU fallback
SQLAlchemy + SQLite	Async lead persistence; Alembic migrations; one env var swap to PostgreSQL
Prometheus	Stage latency histograms, lead score distribution, RAG chunk counts, hallucination counters
LangSmith	Per-agent token usage, latency, prompt + output trace when API key is configured

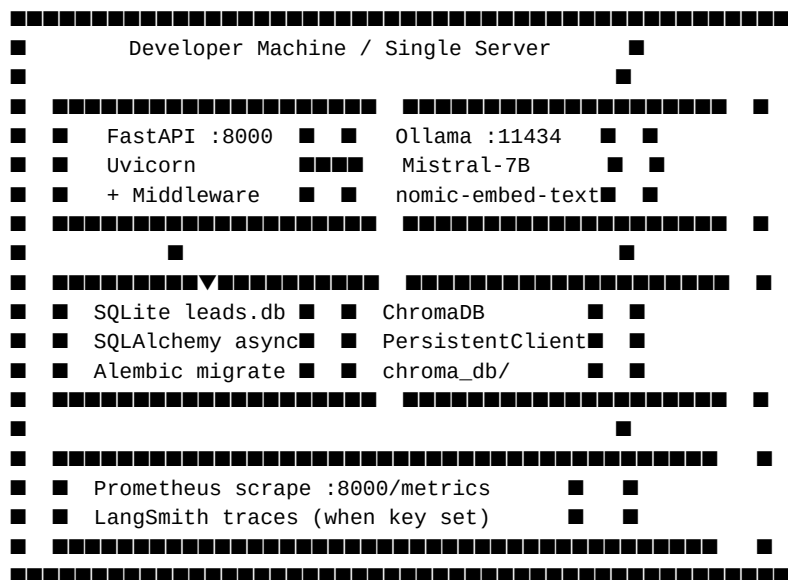
Request Lifecycle

- ① POST /api/v1/search {keyword, location}
-
- ② FastAPI validates → creates initial LeadState → invokes LangGraph
-
- ③ ResearchAgent
 - SerpSearchTool → SerpAPI Google search → top organic result (company URL)
 - ScraperTool → httpx + BeautifulSoup → company page content
 - ContactFinderTool → regex DM title/name extraction → decision_makers[]
 - Mistral-7B → 3-paragraph structured summary
(COMPANY OVERVIEW / HR PAIN POINTS / HRMS FIT SIGNAL)
-
- ④ QualificationAgent
 - score_company_size() 0-20 (employee headcount brackets)
 - score_industry() 0-20 (primary / adjacent / unknown)
 - score_tech_stack() 0-20 (spreadsheet / legacy HRMS / modern HRMS)
 - score_dm_reachability() 0-20 (email+LinkedIn / name / title / none)
 - score_growth_signal() 0-20 (hiring surge / expansion / stable / contracting)
 - completeness_gate < 40% data completeness → disqualify regardless of score
 - Mistral-7B → 2-sentence reasoning (best signal + biggest risk)
 - conditional edge score ≥ 35 → SalesAgent | else → disqualified
-
- ⑤ SalesAgent (qualified leads only)
 - build_rag_query(company, industry, tech_stack, growth_signal, summary)
 - nomic-embed-text → embed query → ChromaDB cosine search → top-5 product chunks
 - Mistral-7B → cold email (150-220 words, RAG-grounded, no hallucinated claims)
 - Mistral-7B → LinkedIn message (≤300 chars, no emojis)
 - SelfCritiqueTool → hallucination check → rag_context_used[] stored in LeadState
-
- ⑥ save_lead() → SQLite
- ⑦ return {lead_id, db_id, lead{score_breakdown, outreach_email, rag_context_used}}

Data Flow

- WRITE path (POST /search):
- keyword / location
 - SerpAPI JSON results
 - scraped HTML → company fields (name, domain, employees, tech_stack, growth_signal)
 - ContactFinder regex → decision_makers[]
 - Mistral-7B → raw_summary (3 paragraphs)
 - 5 deterministic functions → qualification_score + score_breakdown
 - Mistral-7B → qualification_reasoning (2 sentences)
 - ChromaDB cosine retrieval → rag_context_used[]
 - Mistral-7B → outreach_email + linkedin_message
 - SQLite leads table (full LeadState persisted)
- READ path (GET /leads):
- query params (sort_by, status, limit, offset)
 - SQLAlchemy async SELECT → ordered, filtered JSON response
- RAG debug path (GET /rag/query):
- ?q=query string
 - nomic-embed-text embed → ChromaDB cosine search → chunks[] returned

Deployment View



Production upgrade path (zero code changes required):

DATABASE_URL	→	postgresql+asyncpg://...	(one env var)
ChromaDB	→	chromadb HTTP server	(one env var)
Uvicorn	→	uvicorn --workers 4	(one flag)
Ollama	→	shared GPU server	(one env var)

Horizontal scale:

nginx / ALB → [API pod 1 | API pod 2 | API pod 3]

↓

PostgreSQL · Ollama GPU · ChromaDB

LLM Model Selection, Quantization & GPU Requirements

Model: Mistral-7B-Instruct-v0.3 Q4_K_M

Why Mistral-7B over alternatives:

Model	Parameters	Why Considered	Decision
Mistral-7B-Instruct-v0.3	7B	Strong instruction-following, Indian-English well-represented, 8k context, Apache 2.0	Selected
Llama-3-8B-Instruct	8B	Excellent general quality, larger context	Larger memory footprint; Mistral performs equally on structured output tasks
Phi-3-Mini-4k	3.8B	Very small, fast	4k context too short for multi-paragraph research summaries
Gemma-7B	7B	Strong quality	Restrictive usage policy; less instruction-tuned for B2B sales copy

Why 7B is the right size for this workload:

- Research summary: 300–400 token output, 2k input → fits 7B context comfortably
- Cold email: 200–220 words output, constrained by prompt → 7B produces structured output reliably
- Qualification reasoning: 2-sentence output — one model across all 3 agents keeps deployment simple
- Larger models (13B, 34B) require ≥ 24 GB VRAM, eliminating single-GPU deployment

Quantization Strategy: Q4_K_M

Q4_K_M is a 4-bit GGUF quantization format from llama.cpp, implemented in Ollama:

- **Q4**: 4-bit integer weights (from 16-bit float) — 75% memory reduction vs. full precision
- **K**: K-quants method — mixed precision per tensor block; higher-importance tensors retain more bits
- **M**: "Medium" variant — balanced between quality loss and memory savings vs. Q4_K_S or Q4_K_L

Why Q4_K_M specifically:

- Mistral-7B at Q4_K_M: ~4.4 GB VRAM — fits on a single 8 GB GPU (RTX 3070 / 4060 Ti) with headroom for KV cache
- Quality benchmark vs. full FP16: ~97% task performance on instruction-following benchmarks
- Q3 and below show measurable JSON output corruption in structured tasks; Q4_K_M avoids this
- Q5/Q6 would require 6–7 GB VRAM with marginal quality improvement

nomic-embed-text (embedding model): FP16 at ~137M parameters — ~270 MB VRAM, runs concurrently with the LLM.

GPU Requirements

Deployment Target	GPU	VRAM	Notes
Development (single GPU)	RTX 3070 / RTX 4060 Ti	8 GB	Q4_K_M LLM + nomic-embed-text simultaneously
Production (recommended)	RTX 4090 / A10G	24 GB	Higher concurrency; or Q8_0 for better quality
CPU fallback	Any	—	OLLAMA_NUM_GPU=0; ~10× slower; viable for low-volume demos
Cloud GPU	A10G (AWS g5) / T4 (Colab)	16–24 GB	T4 (16 GB) handles Q4_K_M comfortably

Design Decisions

Decision	Choice	Rationale
Orchestration	LangGraph StateGraph	Explicit state machine — edges are code, not LLM decisions. Conditional edge enforces deterministic routing. CrewAI hides agent-to-agent calls behind abstractions

Decision	Choice	Rationale
Scoring	Deterministic functions	5 functions × 0–20 = reproducible, debuggable scores. LLM scoring is inconsistent across runs and hallucinates scores
Vector store	ChromaDB PersistentClient	Zero network dependency vs. HTTP server mode — eliminates one failure surface in local/dev deployment
LLM	Mistral-7B Q4_K_M	4.4 GB VRAM — fits 8 GB GPU with headroom. 97% quality vs. FP16. Apache 2.0 license
Persistence	SQLite → PostgreSQL	Single env var DATABASE_URL swaps storage. Alembic handles schema migration. Zero code change
Prompt management	Jinja2 . j2 templates	Version-controlled, human-readable, testable without LLM. Not string concatenation
Hallucination prevention	3-layer: prompt + citation + critique	Constrained prompts reduce surface; rag_context_used gives full auditability; SelfCritique adds automated verification

Fine-Tuning Strategy

When Fine-Tuning is Better than Prompting

Fine-tuning is NOT the first-line solution for this system. The current prompt-based approach with RAG is preferred because:

- **Prompting + RAG** handles new product features by re-ingesting the website — no model retraining needed
- **Fine-tuning** is appropriate when: (a) output format must be extremely consistent (always valid JSON without post-processing), (b) domain-specific vocabulary is rare in the base model (Indian compliance terms: CLRA, ESIC Form 5A, PT returns), or (c) prompt budget needs to be reduced by baking knowledge into weights

Trigger points where fine-tuning would be justified:

- LLM consistently produces malformed SUBJECT:/BODY: splits requiring extra parsing
- Qualification reasoning hallucinates specific compliance regulations not in RAG context
- Cold emails include generic placeholder language despite specific RAG context

Fine-Tuning Approach: QLoRA with Unsloth

Framework: [Unsloth](<https://github.com/unslothai/unsloth>) + QLoRA (Quantised Low-Rank Adaptation)

Why QLoRA over full fine-tuning:

- Full fine-tuning of 7B parameters requires ~80–120 GB VRAM (impractical)

- QLoRA freezes base model weights and trains small low-rank adapter matrices on top of 4-bit quantized base
- Training VRAM: ~8–12 GB for 7B model — single GPU accessible
- Unsloth accelerates QLoRA training 2–3× over HuggingFace PEFT via custom CUDA kernels

Target modules for LoRA adaptation:

```
target_modules = ["q_proj", "k_proj", "v_proj", "o_proj",
                  "gate_proj", "up_proj", "down_proj"]
lora_r = 16          # rank – higher = more expressive adapter
lora_alpha = 32      # scaling factor
lora_dropout = 0.05
```

Dataset Format

Fine-tuning dataset: JSONL file of {instruction, input, output} triples in Alpaca format:

```
{"instruction": "Write a cold email for a B2B HRMS sale.", "input": "Company: Bharat Forge (Manufacturing)", "output": "Cold email draft"}
{"instruction": "Write a 2-sentence qualification reasoning.", "input": "Company: Metropolis Healthcare", "output": "Qualification reasoning text"}
```

Constructing the training set:

- Run pipeline in live mode on 200–300 target companies (SerpAPI required)
- Human sales expert reviews and edits outputs → ground truth labels
- Add 50 adversarial examples (micro companies, irrelevant industries) → teach graceful decline
- Target: 300–500 high-quality examples (quality >> quantity for instruction fine-tuning)

Training configuration (Unsloth):

```
from unsloth import FastLanguageModel

model, tokenizer = FastLanguageModel.from_pretrained(
    "mistralai/Mistral-7B-Instruct-v0.3",
    max_seq_length=4096,
    load_in_4bit=True,
)
model = FastLanguageModel.get_peft_model(
    model,
    r=16, lora_alpha=32, lora_dropout=0.05,
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj",
                   "gate_proj", "up_proj", "down_proj"],
    bias="none",
    use_gradient_checkpointing="unsloth",
)
```

After training: Export LoRA adapter → merge into base weights → convert to GGUF Q4_K_M → deploy via Ollama (drop-in replacement, same API).

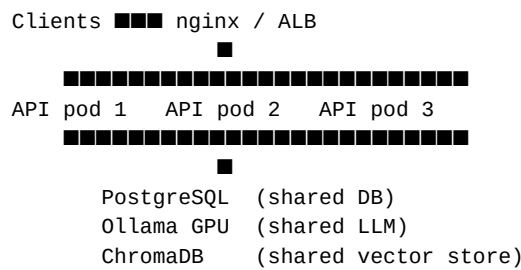
Scaling Approach

Vertical Scaling

Bottleneck	Current	Scaled
LLM throughput	1 Ollama instance, sequential	--num-parallel 2 flag on Ollama

Bottleneck	Current	Scaled
DB write contention	SQLite async	DATABASE_URL → PostgreSQL (asyncpg) — one env var
API concurrency	Uvicorn 1 worker	uvicorn --workers 4 or Gunicorn + Uvicorn workers

Horizontal Scaling



Async pipeline offload (production pattern):

- Move `pipeline.invoke()` off the HTTP thread to a Celery/RQ worker
- API returns `{task_id}` immediately; client polls `GET /leads/{lead_id}` until `status != pending`
- Enables burst handling and prevents HTTP timeouts on slow LLM responses

Observability & Monitoring

Signal	Instrument	Tool	Alert Condition
Latency	stage_latency histogram per API path	Prometheus	p99 > 30s on /api/v1/search
Pipeline throughput	leads_processed_total counter by status	Prometheus	Drop > 50% vs. 5-min baseline
Qualification score	lead_qualification_score histogram	Prometheus	Mean drops below 30
RAG chunk retrieval	rag_chunks_retrieved histogram	Prometheus	p50 = 0 (ChromaDB down or empty)
Hallucinations	hallucination_checks_total by is_grounded	Prometheus	is_grounded=false rate > 10% in 15 min
LLM availability	Exception rate + errors[] in LeadState	Log aggregation	Error rate > 5%
Agent failures	LangSmith traces + errors[] in LeadState	LangSmith dashboard	Any trace with status=error
API failures	HTTP 4xx/5xx via middleware	Prometheus	5xx rate > 1% over 5 min

Hallucination Prevention — 3 Layers

- **Prompt-level constraint** (synchronous): Jinja2 templates explicitly forbid facts not in RAG context — reduces hallucination surface before generation

- **RAG grounding citation** (synchronous): `rag_context_used` stored in every lead — evaluators can verify any claim against cited chunks
- **Automated self-critique** (async): `SelfCritiqueTool` runs a second LLM pass comparing email claims against RAG chunks — emits `hallucination_checks_total{is_grounding=...}` Prometheus counter

LangSmith Agent Tracing

When `LANGCHAIN_API_KEY` is set, every `LangGraph` node is traced in LangSmith:

- Per-agent token usage, latency, prompt content, and output visible in timeline view
- Failed nodes surface as error spans with full stack trace
- Trace comparison across runs identifies prompt regression after model or template updates